

Assignment – 4.1

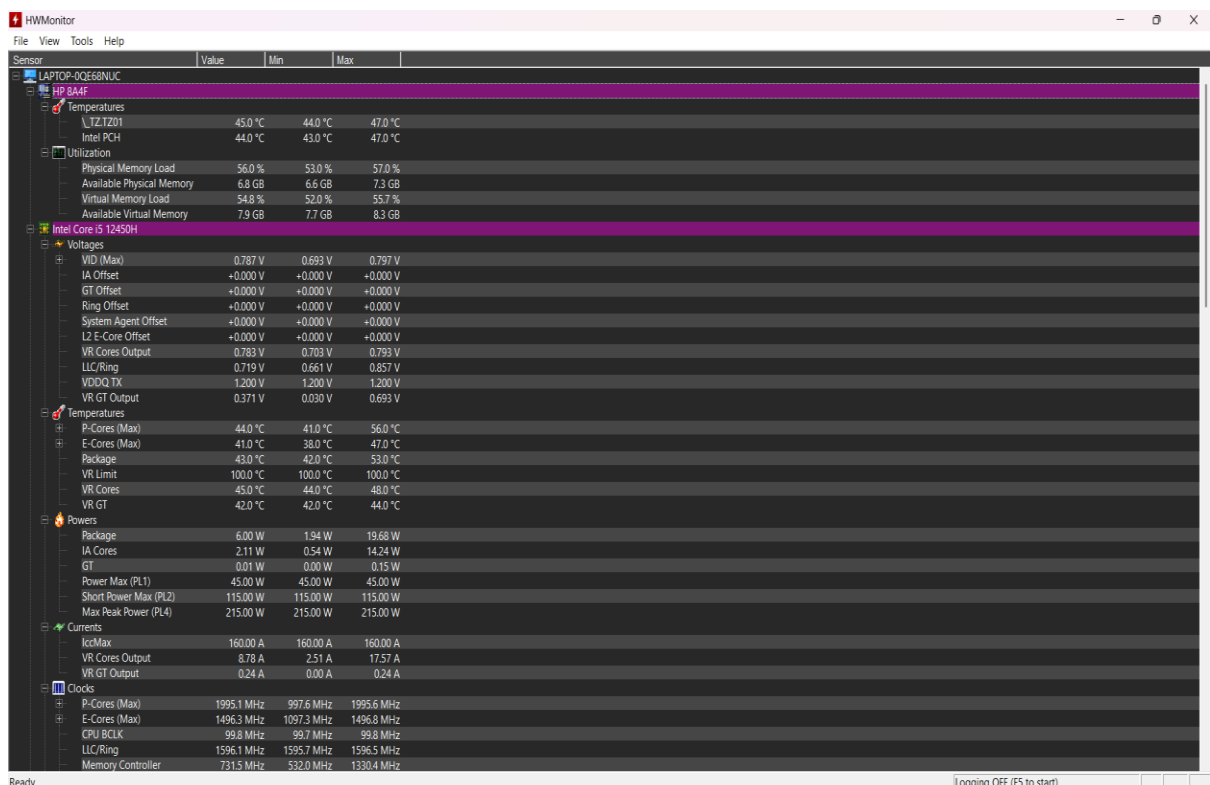
Q - 1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

No.	Real-life scenario	How the multimeter helps
1.	Computer/PC not turning on	Checks if the power supply is giving the correct voltage.
2.	Laptop battery not charging	Checks the battery and charger voltage.
3.	Smartphone not charging	Checks the charging port and connections for faults.

Q - 2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Hint: Focus on the codes shown and how they relate to hardware issues.

- The POST card displays **codes during the computer's startup (POST) process**, showing which hardware is being tested.
- In the demo, codes such as **3387, 2900, and AA00** are displayed as the BIOS progresses through its checks.
- If the card stops at a particular code, it can help identify a **hardware problem such as RAM, CPU, or motherboard failure**.
- In the demonstrated system, the final **AA00 code indicated successful startup and OS loading**.

Q - 3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Constraint: Blur or crop out any personal information before submitting.



Q - 4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Checks voltage, current, and resistance	Faulty power supply or damaged charging cable
POST Card	Checks motherboard startup/POST codes	RAM or motherboard failure
Software Diagnostics	Checks system hardware and software health	Hard disk/SSD errors or high CPU temperature